Kernix 0.11.1 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

35 workloads

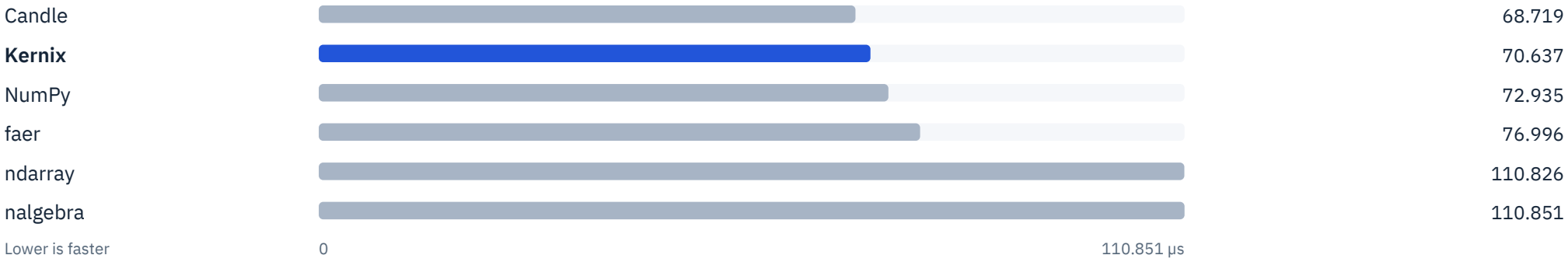
119 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



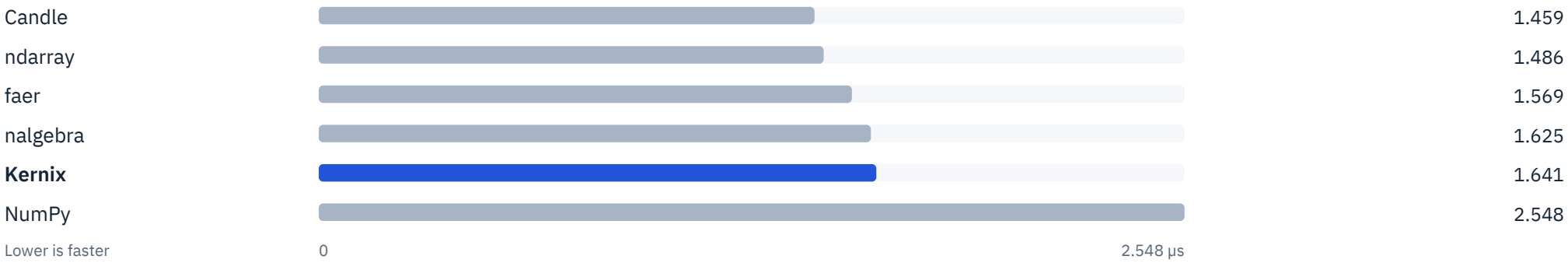
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

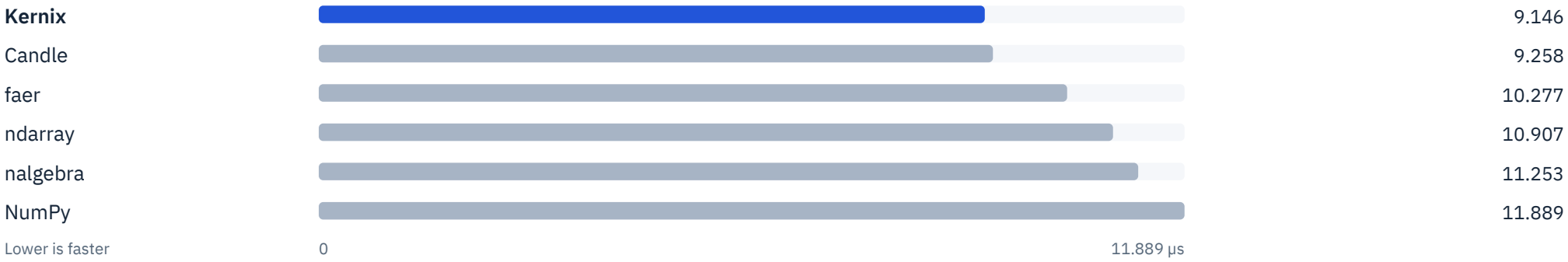
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



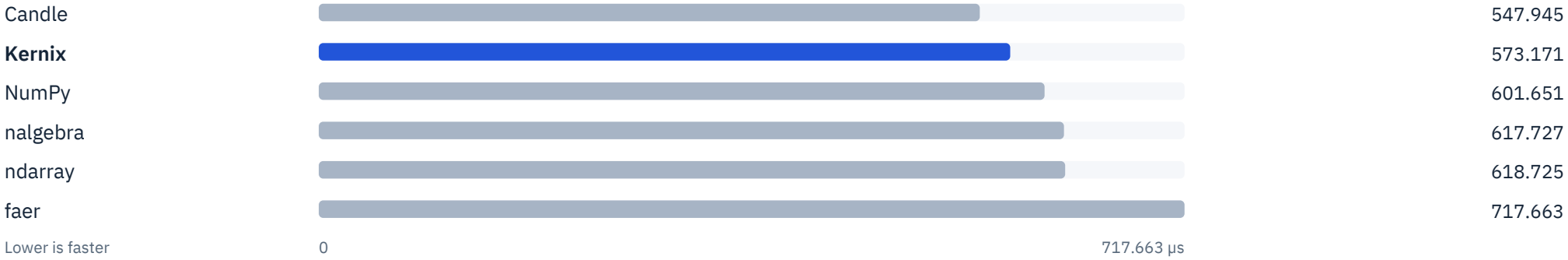
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

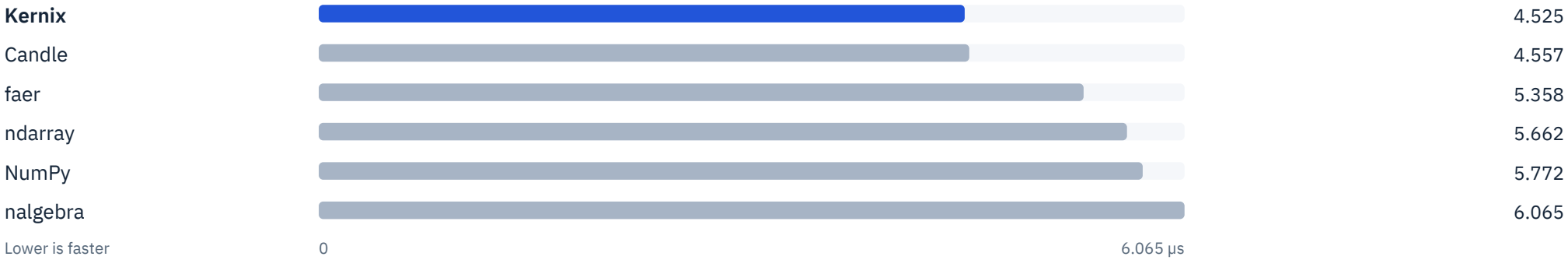
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



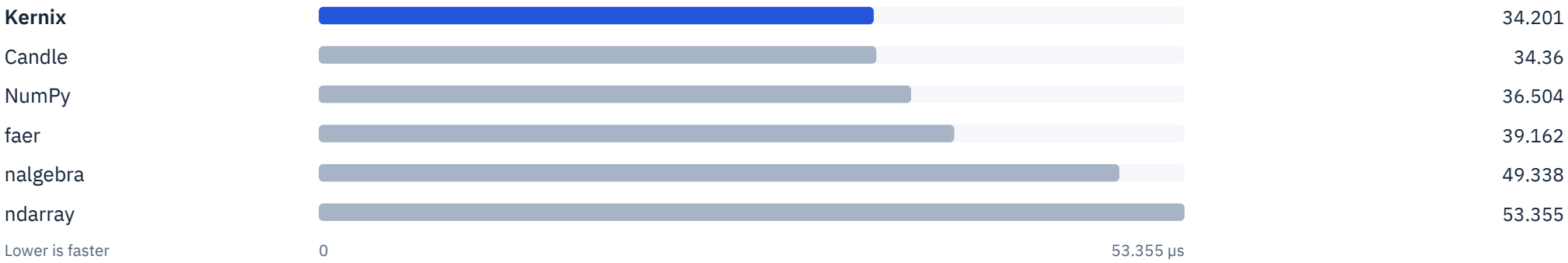
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

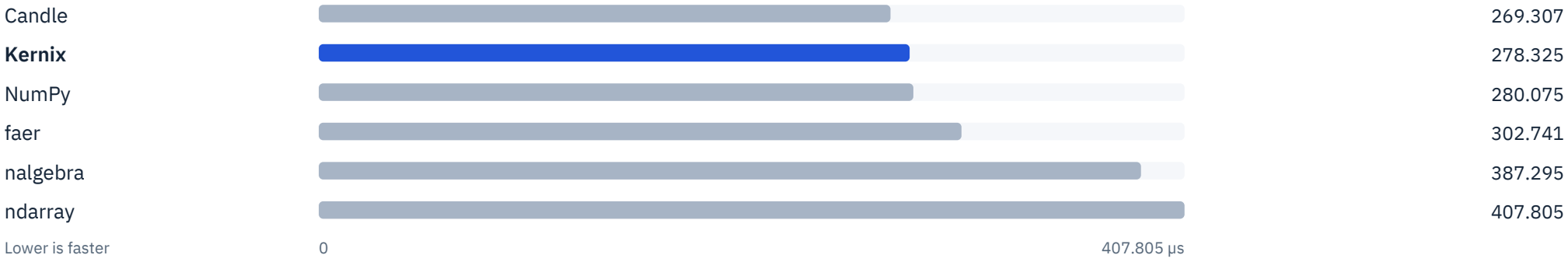
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



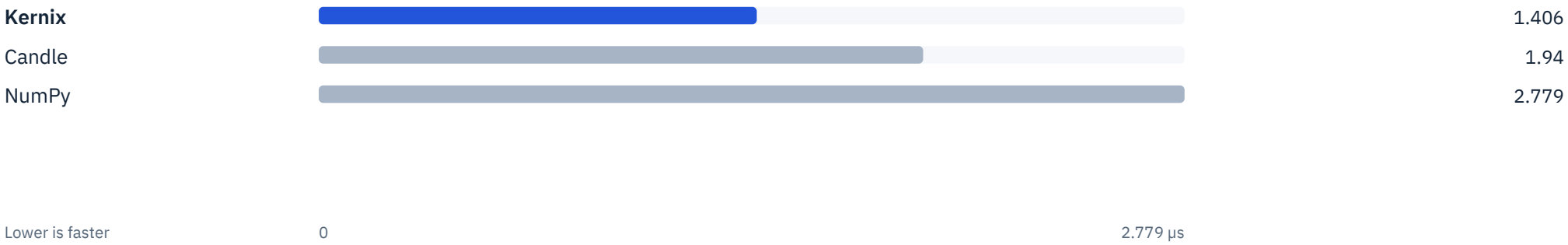
Kernix 0.11.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

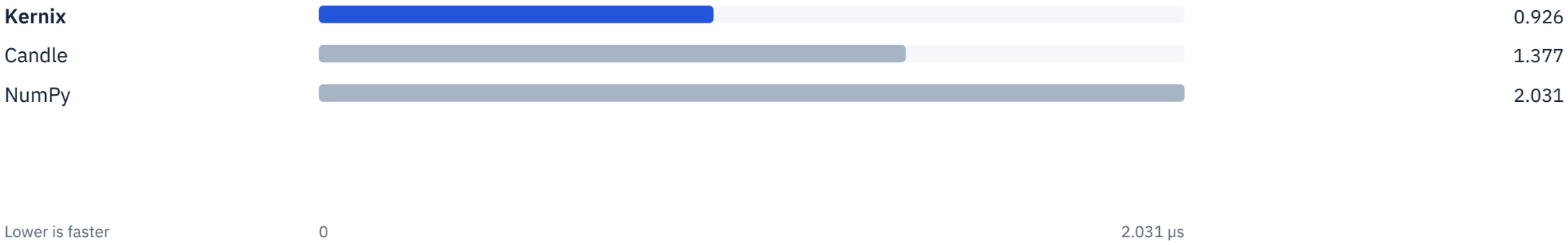
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

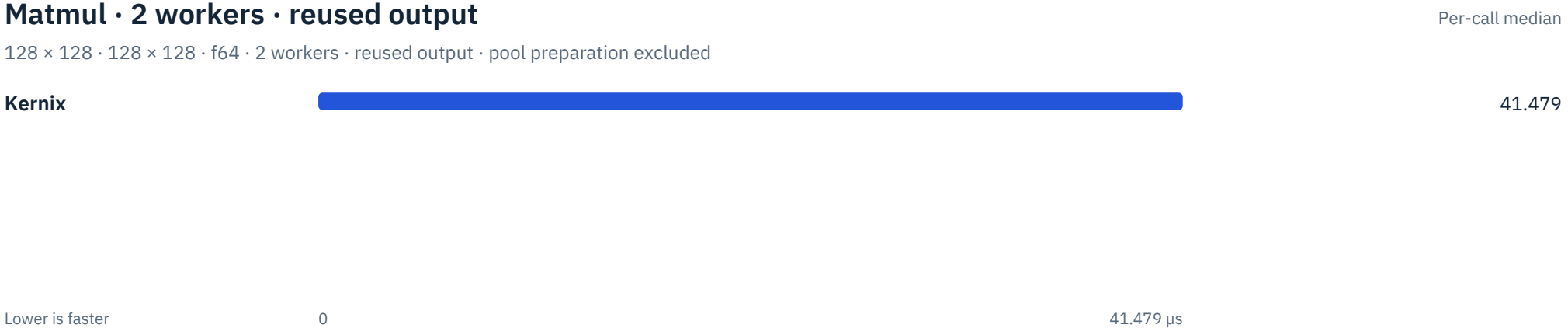
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



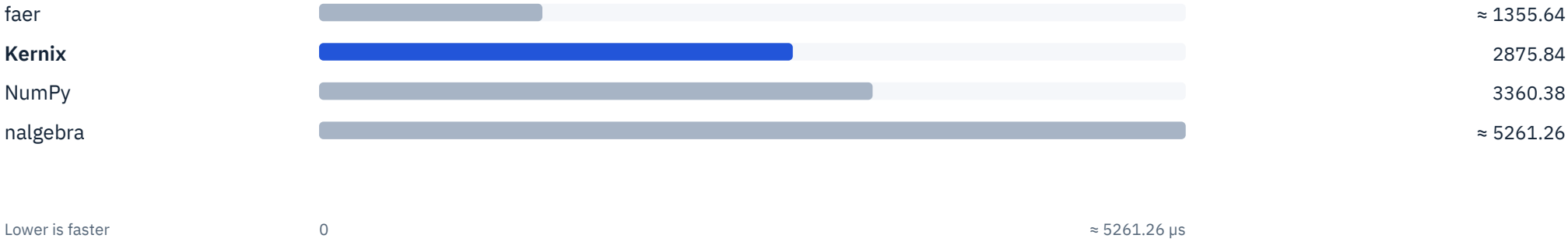
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



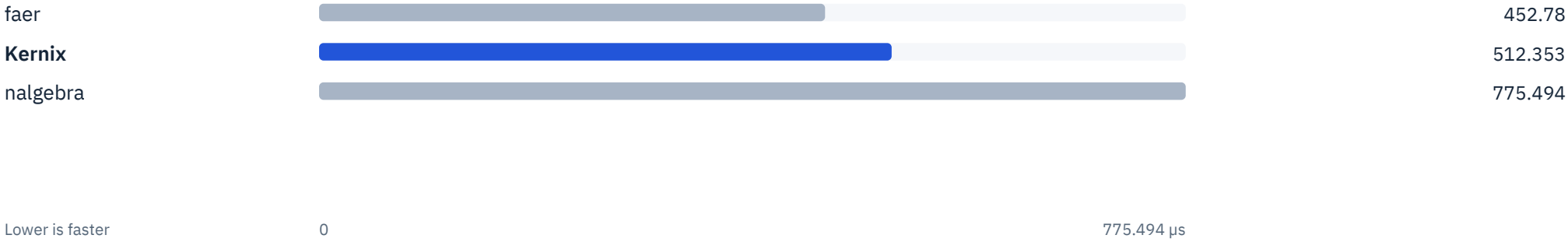
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

Per-call median

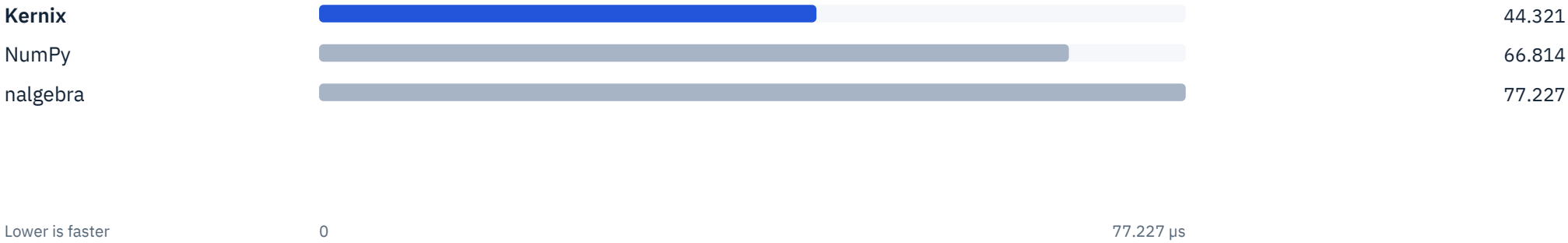
256 × 64 · f64 · 1 worker · allocated output



Minimum-norm least squares

Per-call median

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



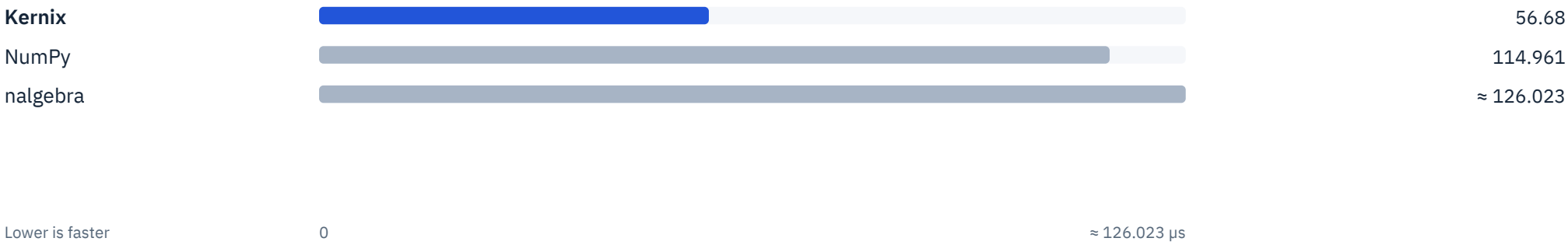
Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

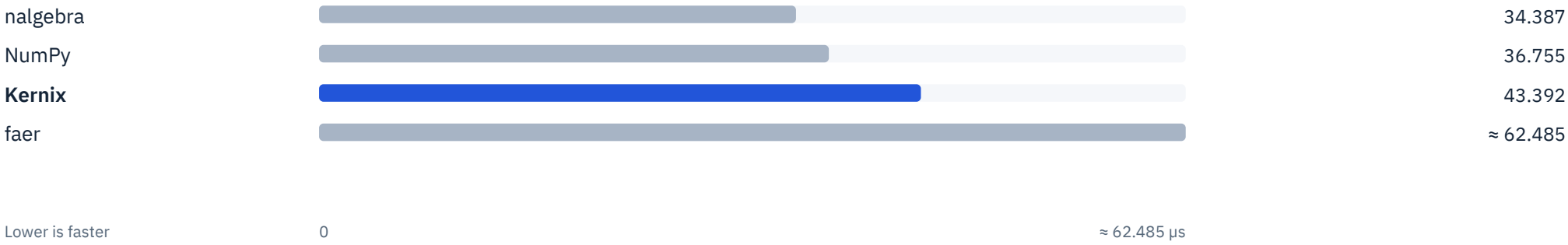
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



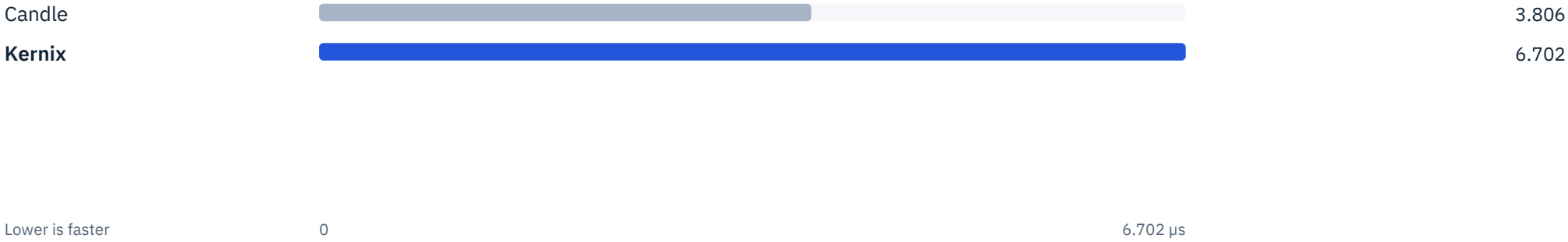
Kernix 0.11.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented

Per-call median



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented

Per-call median



Kernix 0.11.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



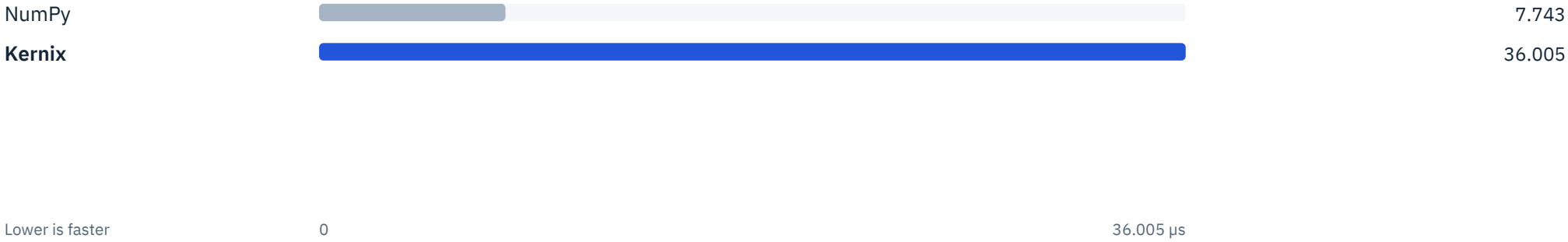
Kernix 0.11.1 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



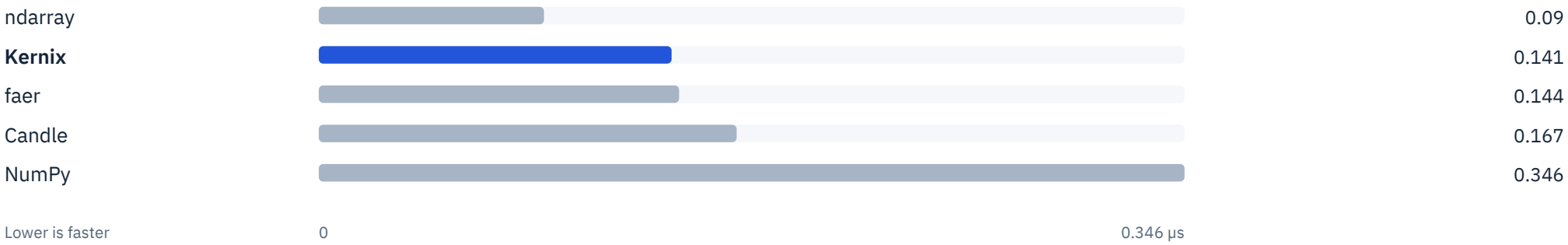
Kernix 0.11.1 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9904882688649335	0.9548349739432542 – 1.0424050632911392

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.012244500410469	0.996820129959906 – 1.0277707380187007

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0019341832842976	0.9540607054963085 – 1.0580849852211405

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9959523669853934	0.9901368615535494 – 1.0083672209329557

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9991795672844394	0.9917177400090559 – 1.0197192236299086

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9321063394683027	0.9020817270624518 – 0.9764801297648013

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9973845772650938	0.9513612643104681 – 1.0198856125019133

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9919104759336659	0.971837475943946 – 1.0259382052210737

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7574906367041199	0.7263590680676107 – 0.8196382428940568

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7099767981438515	0.6951219512195121 – 0.7725225225225225

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9840000693012699	0.9571668921497207 – 1.0007533641606483

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9955355668450756	0.9890255559008289 – 1.0107981826317975

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0037438332284923	0.9868867083576638 – 1.0199812625802072

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9707103267549454	0.5377926771777277 – 0.9987317073170732

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

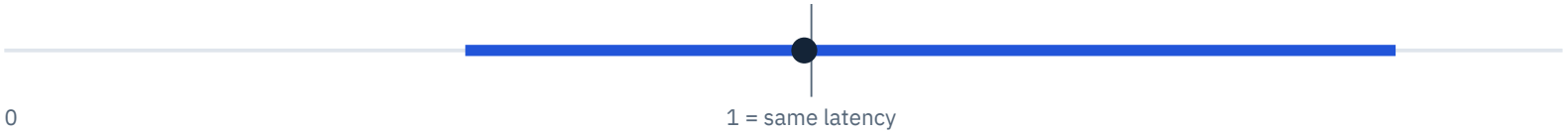


CURRENT / BASELINE	95% INTERVAL
0.9909823528068854	0.9644791506637074 – 1.0365048026342756

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9912084047650446	0.5711984041736996 – 1.7238628349788434

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	73137	73.137
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	69083	69.083
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	39283	39.283
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	34252	34.252
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1459	1.459
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1486	1.486
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1625	1.625
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1569	1.569
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2548	2.548
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1641	1.641

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10907	10.907
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	11889	11.889
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9258	9.258
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10277	10.277
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11253	11.253
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9146	9.146
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	72935	72.935
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	110826	110.826
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	70637	70.637
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	76996	76.996

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	110851	110.851
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	68719	68.719
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	601651	601.651
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	547945	547.945
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	617727	617.727
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	618725	618.725
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	717663	717.663
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	573171	573.171
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	5662	5.662
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	5772	5.772

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5358	5.358
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4525	4.525
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4557	4.557
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6065	6.065
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	49338	49.338
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	39162	39.162
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	34360	34.36
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	53355	53.355
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	34201	34.201
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	36504	36.504

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	387295	387.295
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	280075	280.075
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	407805	407.805
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	302741	302.741
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	278325	278.325
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	269307	269.307
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1940	1.94
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2779	2.779
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1406	1.406
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1377	1.377

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2031	2.031
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	926	0.926
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	9501	9.501
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	41479	41.479
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	297428	297.428
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	22180	22.18
Cholesky factorization / nalgebra 512 × 512 · f64	5261263	5261.263
Cholesky factorization / NumPy 512 × 512 · f64	3360380	3360.38
Cholesky factorization / faer 512 × 512 · f64	1355642	1355.642
Cholesky factorization / Kernix 512 × 512 · f64	2875840	2875.84

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	71704	71.704
Column-pivoted QR / nalgebra 96 × 32 · f64	76393	76.393
Column-pivoted QR / Kernix 96 × 32 · f64	56430	56.43
Column-pivoted QR / nalgebra 256 × 64 · f64	775494	775.494
Column-pivoted QR / faer 256 × 64 · f64	452780	452.78
Column-pivoted QR / Kernix 256 × 64 · f64	512353	512.353
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	77227	77.227
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	66814	66.814
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	44321	44.321
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	114961	114.961

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	56680	56.68
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	126023.00000000001	126.02300000000001
Thin QR / faer 96 × 32 · f64	62484.99999999999	62.48499999999999
Thin QR / NumPy 96 × 32 · f64	36755	36.755
Thin QR / Kernix 96 × 32 · f64	43392	43.392
Thin QR / nalgebra 96 × 32 · f64	34387	34.387
Thin QR / faer 256 × 64 · f64	394945	394.945
Thin QR / nalgebra 256 × 64 · f64	374716	374.716
Thin QR / Kernix 256 × 64 · f64	469928	469.928
Thin QR / NumPy 256 × 64 · f64	338736	338.736

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	18925	18.925
Cholesky factorization / faer 64 × 64 · f64	8439	8.439
Cholesky factorization / NumPy 64 × 64 · f64	10374	10.374
Cholesky factorization / nalgebra 64 × 64 · f64	13118	13.118
Cholesky factorization / NumPy 128 × 128 · f64	73702	73.702
Cholesky factorization / nalgebra 128 × 128 · f64	89792	89.792
Cholesky factorization / faer 128 × 128 · f64	37553	37.553
Cholesky factorization / Kernix 128 × 128 · f64	110399	110.399
Cholesky factorization / NumPy 256 × 256 · f64	516323	516.323
Cholesky factorization / faer 256 × 256 · f64	192757	192.757

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	639934	639.934
Cholesky factorization / Kernix 256 × 256 · f64	495698.00000000006	495.69800000000006
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3806	3.806
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6702	6.702
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	3983	3.983
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2228	2.228
Model inference / Candle batched normalized/sliced MLP · f64	3152	3.152
Model inference / Kernix batched normalized/sliced MLP · f64	2151	2.151
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2945	2.945
Model recording + forward / Candle batched normalized/sliced MLP · f64	5180	5.18

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	2973	2.973
Model backward / Candle prepared graph → fresh gradients · f64	14581	14.581
Parameter update / Kernix transactional SGD, fixed gradients · f64	515	0.515
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	36005	36.005
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	7743	7.743
Row softmax / Candle 256 × 64 · f64	81850	81.85
Row softmax / ndarray 256 × 64 · f64	68484	68.484
Row softmax / Kernix 256 × 64 · f64	81343	81.343
Row softmax / NumPy 256 × 64 · f64	97519	97.519
Create zeros / faer 64 × 64 · f32	144	0.144

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	167	0.167
Create zeros / NumPy 64 × 64 · f32	346	0.346
Create zeros / Kernix 64 × 64 · f32	141	0.141
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create ones / Candle 64 × 64 · f32	236	0.236
Create ones / ndarray 64 × 64 · f32	140	0.14
Create ones / NumPy 64 × 64 · f32	1213	1.213
Create ones / Kernix 64 × 64 · f32	212	0.212
Create ones / faer 64 × 64 · f32	182	0.182